SHUBHAM SINGHAL

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EDUCATION

Columbia University in the City of New York — New York, NY
Master of Science in Computer Science — *Dec 2018*

IIIT Delhi — Delhi, India
Bachelor of Technology in Computer Science — *May 2014*

SKILLS & INTERESTS

Programming Languages	Java, Python, C#
Tools and Technologies	REST APIs, Microservices, LangGraph, SQL, Cassandra, Argo CI/CD, Azure, AWS

PROFESSIONAL EXPERIENCE

Adobe Inc., *Sr. Software Engineer* | San Jose, CA — March 2023 – Present

Multi-Agent Observability Platform for Commerce Systems

- Architected and leading the development of an agentic observability platform spanning ordering, payments, and fraud domains, enabling end-to-end transaction tracing, monitoring and cross-service state reconciliation across microservices
- Designed and implemented a LangGraph-based orchestration system with iterative reasoning and dynamic agent routing to power reliable autonomous diagnostics. Integrated MCP connectors to Databricks, Splunk, Jira, and internal runbooks, enabling agents to correlate logs, metrics, tickets, and operational knowledge for faster and more accurate root-cause analysis
- Productionized the platform with reliability guardrails like confidence thresholds, cross-source validation, and human escalation workflows, driving adoption across commerce engineering and improving L1/L2 support efficiency

Payments Platform Optimization

- Designed and implemented a recurring billing retry framework aligned with data integrity mandates that led to a 2% improvement in global authorization rates
- Led the shift to a configuration-driven payment integration framework, separating business logic from transaction flows and enabling dynamic, config-based orchestration streamlining vendor onboarding and accelerating compliance

Meta Platforms, *Software Engineer* | Menlo Park, CA — Sep 2021 – January 2023

Creator Monetization Infrastructure

- Led the design and development of a framework to identify and resolve ghost subscriptions (where subscribers continue paying to demonetized/deleted creators without receiving exclusive content). Built a new pub/sub messaging queue to enable efficient event handling. This reliability effort led to the identification of nearly 400k ghost subscriptions worth $900k in refunds

Creator Payout Transparency

- Developed and designed payments backend infrastructure and payout APIs to surface earning metrics for subscriptions/stars purchases, incentive bonuses and allow for manual adjustments which enabled creator payout transparency.
- Implemented a payments reconciliation system to monitor divergence between in-app purchase reports (iOS and Android) and Meta's payments ledger. Identification and reconciliation of inconsistent transactions led to 18% reduction in payments engine's discrepancies.

Microsoft, *Software Engineer 2* | Redmond, WA — Feb 2019 – Sep 2021

Device-License Entitlement Verification using Blockchain

- Architected and implemented smart contracts on Azure Blockchain to immutably record device-license entitlements for Microsoft IP protection, and built serverless APIs (Azure Functions, Logic Apps) to securely integrate consortium ledger data with enterprise systems

Online Marketplace Piracy Detection

- Built a framework to identify fraudulent listings on 3rd party online marketplaces in sanctioned regions. Since the listings were multilingual, this was achieved by developing a language agnostic translation service to retrieve listing description and seller information. Nearly 15k listings worth $340k were flagged across 8 sanctioned markets.

Google, *Software Engineering Intern* | Mountain View, CA — May 2018 – Aug 2018

Orientation Estimator (Location Context, Maps)

- Improved orientation performance for low-end mobile devices by using crowd-sourced sensor traces and augmenting it with artificially infused impairments. Designed and implemented metrics to validate these improvements for Google Maps and Geo-AR.